List Vs Set Vs Tuple Vs Dictionary in Python

Feature	List	Set	Tuple	Dictionary
Definition	Ordered collection of items	Unordered collection of unique items	Ordered, immutable collection of items	Collection of key-value pairs
Syntax	list1 = [10, 20, 30]	set1 = {10, 20, 30}	tuple1 = (10, 20, 30)	dict1 = {"a": 1, "b": 2}
Order maintained?	✅ Yes	❌ No (unordered)	✅ Yes	✅ Yes (since Python 3.7+)
Indexing supported?	✅ Yes (list1[0])	❌ No	✅ Yes (tuple1[1])	✅ Keys can be accessed (dict1["a"])
Mutable (changeable)?	✅ Yes	✅ Yes (can add/remove items)	❌ No (immutable)	✅ Yes
Duplicates allowed?	✅ Yes	❌ No (all items must be unique)	✅ Yes	❌ Keys must be unique (values can repeat)
Heterogeneous data allowed?	✅ Yes	✅ Yes	✅ Yes	✅ Yes
Common use cases	Store multiple values in order, can modify	Store unique values, remove duplicates, fast membership test	Fixed collection of items (like constants)	Store data as key-value pairs (like a real-world dictionary)
Example Operations	append(), remove(), sort()	add(), remove(), union(), intersection()	Count, index (tuple.count(), tuple.index())	keys(), values(), items(), update()

In simple words:

- **List** → Ordered, changeable(mutable), allows duplicates.
- **Set** → Unordered, unique values only, mutable.
- **Tuple** → Ordered, unchangeable(immutable), allows duplicates.
- **Dictionary** → Stores data in **key-value** pairs, keys must be unique.